Q1. Two valid one-electron wavefunctions are given below:

$$(i)\ \psi_{n,l,m_l}(r,\theta,\phi) = R_{3,0}(r)Y_{0,0}(\theta,\phi) \qquad (ii)\ \psi_{n,l,m_l}(r,\theta,\phi) = R_{3,2}(r)Y_{2,-1}(\theta,\phi)$$

What are the quantum numbers for each of the wavefunctions?

A1

$\Psi_{n,l,m_l}(r,\theta,\phi) = R_{3,0}(r)\, Y_{0,0}(\theta,\phi)$

n Principal quantum no. 3

l angular momentum quantum no. 0

m_l magnetic quantum no. 0

$\Psi_{n,l,m_l}(r,\theta,\phi) = R_{3,2}(r)\, Y_{2,-1}(\theta,\phi)$

n : 3

l : 2

m_l : -1

Q2. The ψ_{2s} atomic orbital for hydrogen-like atom is given as:

$$(i)\ \psi_{2s} = \frac{1}{4\sqrt{2\pi}}\left(\frac{Z}{a_0}\right)^{3/2}\left(2 - \frac{Zr}{a_0}\right)e^{-Zr/2a_0}$$

How many radial node do you expect to see for ψ_{2s}? Find, where the radial node of the wavefunction occurs in terms of a_0 for H, He^+, Li^{2+} atoms, and comment on your results.

A2 No. of radial nodes $= (n - l - 1)$

$= (2 - 0 - 1)$

$= 1$

At the radial node $\psi_{2s} = 0$

$$\frac{1}{4\sqrt{2\pi}}\left(\frac{Z}{a_0}\right)^{3/2}\underbrace{\left(2 - \frac{Zr}{a_0}\right)}_{0}e^{-Zr/2a_0} = 0$$

Let's make $2 - \frac{Zr}{a_0} = 0$

$$\Rightarrow r = \frac{2a_0}{Z}$$

For H, $Z = 1$, $r = 2a_0$

For He^+, $Z = 2$, $r = \frac{2a_0}{2} = a_0$

For Li^{2+}, $Z = 3$, $r = \frac{2a_0}{3}$

$r_H > r_{He^+} > r_{Li^{2+}}$

Atomic no. increases ↑

Radius decreases ↓

Q3. A pair of $2p$ atomic orbitals are given as follows:

$$\psi_{2,1,\pm1} = \frac{1}{\sqrt{64\pi}}\left(\frac{Z}{a_0}\right)^{3/2}\left(\frac{Zr}{a_0}\right)e^{-Zr/2a_0}\ \sin\theta\ \mathrm{e}^{\pm i\phi}$$

Show the following linear combinations are real:

$$(a)\quad \psi_{2p_x} = \frac{1}{\sqrt{2}}\left(\psi_{2,1,+1} + \psi_{2,1,-1}\right), \qquad (b)\quad \psi_{2p_y} = \frac{1}{\sqrt{2}\,i}\left(\psi_{2,1,+1} - \psi_{2,1,-1}\right)$$

A3

$$\psi_+ = k\ r\ e^{-\frac{Zr}{2a_0}}\ \sin\theta\ e^{+i\phi}$$

$$\psi_- = k\ r\ e^{-\frac{Zr}{2a_0}}\ \sin\theta\ e^{-i\phi}$$

$$\frac{1}{\sqrt{64\pi}}\left(\frac{Z}{a_0}\right)^{3/2}\left(\frac{Z}{a_0}\right)e^{-Zr/2a_0}$$

$$= \frac{1}{\sqrt{64\pi}}\left(\frac{Z}{a_0}\right)^{5/2}e^{-Zr/2a_0}$$

$$k = \frac{1}{8\sqrt{\pi}}\left(\frac{Z}{a_0}\right)^{5/2}e^{-Zr/2a_0}$$

$$\psi_{2p_x} = \frac{1}{\sqrt{2}}\left(\psi_+ + \psi_-\right)$$

$$= k\frac{1}{\sqrt{2}}\, r\sin\theta\left(e^{+i\phi} + e^{-i\phi}\right)$$

$$= k\frac{1}{\sqrt{2}}\, r\sin\theta\left(\cos\phi + i\sin\phi + \cos\phi - i\sin\phi\right)$$

$$= k\frac{1}{\sqrt{2}}\, r\sin\theta \times 2\cos\phi$$

$$= k\frac{1}{\sqrt{2}}\, 2 \times \underbrace{r\sin\theta\cos\phi}_{x \text{ in spherical polar coordinates}}$$

$$= \frac{1}{4\sqrt{\pi}}\left(\frac{Z}{a_0}\right)^{3/2} e^{-Zr/2a_0}\,\frac{1}{\sqrt{2}} \times 2 \times r\sin\theta\cos\phi$$

$$= \frac{1}{4\sqrt{\pi}\times\sqrt{2}}\left(\frac{Z}{a_0}\right)^{3/2} e^{-Zr/2a_0} \times r\sin\theta\cos\phi$$

$$= \frac{1}{\sqrt{32}}\frac{1}{\sqrt{\pi}}\left(\frac{Z}{a_0}\right)^{3/2}\left(\frac{Zr}{a_0}\right) e^{-Zr/2a_0} \times \sin\theta\cos\phi$$

$$= \frac{1}{\sqrt{32\pi}}\left(\frac{Z}{a_0}\right)^{3/2}\left(\frac{Zr}{a_0}\right) e^{-Zr/2a_0}\sin\theta\cos\phi$$

$$\boxed{\psi_{2p_x} = \frac{1}{\sqrt{32\pi}}\left(\frac{Z}{a_0}\right)^{3/2}\left(\frac{Zr}{a_0}\right) e^{-Zr/2a_0}\sin\theta\cos\phi}$$

$$\psi_{2p_y} = \frac{1}{\sqrt{2}\,i}\left(\psi_+ - \psi_-\right)$$

$$= k\,\frac{1}{\sqrt{2}\,i}\left(r\sin\theta\, e^{+i\phi} - r\sin\theta\, e^{-i\phi}\right)$$

$$= k\,\frac{1}{\sqrt{2}\,i}\, r\sin\theta\left(\cancel{\cos\phi} + i\sin\phi - \cancel{\cos\phi} + i\sin\phi\right)$$

$$= k\,\frac{1}{\sqrt{2}\,\cancel{i}}\, r\sin\theta \times 2\cancel{i}\sin\phi$$

$$= \frac{1}{4\cancel{8}\sqrt{\pi}}\left(\frac{Z}{a_0}\right)^{3/2}\left(\frac{Z}{a_0}\right)\frac{2}{\sqrt{2}}\, e^{-Zr/2a_0}\, \underbrace{r\sin\theta\sin\phi}_{\text{"}y\text{" in spherical Polar coordinates!}}$$

$$= \frac{1}{\sqrt{32\pi}}\left(\frac{Z}{a_0}\right)^{3/2}\left(\frac{Zr}{a_0}\right) e^{-Zr/2a_0}\sin\theta\sin\phi$$

$$\boxed{\psi_{2p_y} = \frac{1}{\sqrt{32\pi}}\left(\frac{Z}{a_0}\right)^{3/2}\left(\frac{Zr}{a_0}\right) e^{-\frac{Zr}{2a_0}}\sin\theta\sin\phi}$$

Q4. The ψ_{1s} atomic orbital of a hydrogen is given as:

$$\psi_{1s} = \frac{2}{a_0^{3/2}} e^{-r/a_0}$$

Calculate at what distance from the nucleus the 1s-electron of the hydrogen atom can be found with the maximum radial probability density, $r^2\,|\psi_{1s}|^2$.

Ans

Radial Probability Density $\boxed{P(r) = r^2 |\psi|^2}$

$$P(r) = r^2 \frac{4}{a_0^3} e^{-2r/a_0}$$

$$= \frac{4}{a_0^3} r^2 e^{-2r/a_0} \quad \ldots\ldots\ldots \text{Eq. 1}$$

By differentiating

$$\frac{dP(r)}{dr} = \left(r^2 \times e^{-\frac{2r}{a_0}} \times \left(\frac{-2}{a_0}\right) + e^{-\frac{2r}{a_0}} \times 2r \right)$$

$$= r \times e^{-\frac{2r}{a_0}} \left[-r \times \left(\frac{2}{a_0}\right) + 2 \right]$$

$$= r\, e^{-2r/a_0} \left[2 - 2r/a_0 \right]$$

From the maxima - minima condition,

$$\left.\frac{dP(r)}{dr}\right|_{r=r_{max}} = 0 \;//\; 2 - 2r/a_0 = 0 \Rightarrow r_{max} \times \frac{2}{a_0} = 2$$

$$\Rightarrow \boxed{r_{max} = a_0}$$

** FOR REF. STUDY

PROOF FOR MAXIMA

$$\frac{dP(r)}{dr} = \left(r^2 \times e^{-\frac{2r}{a_0}} \times \left(\frac{-2}{a_0}\right) + e^{-\frac{2r}{a_0}} \times 2r \right)$$

By differentiating

$$\Rightarrow \frac{d^2P(r)}{dr} =$$

$$\left(\frac{-2}{a_0}\right)\left(r^2 \times e^{-\frac{2r}{a_0}} \times \left(\frac{-2}{a_0}\right) + e^{-\frac{2r}{a_0}} \times 2r \right)$$

$$+ \left(e^{-\frac{2r}{a_0}} \times 2 + 2r \times e^{-\frac{2r}{a_0}} \times \left(\frac{-2}{a_0}\right) \right)$$

$$= \left[r^2 \left(\frac{-2}{a_0}\right)^2 + 2r\left(\frac{-2}{a_0}\right) + 2 + 2r \times \left(\frac{-2}{a_0}\right) \right] \times e^{\frac{-2r}{a_0}}$$

$$= \left[\left(\frac{-2}{a_0}\right) \left\{ \left(\frac{-2}{a_0}\right) r^2 + 4r \right\} + 2 \right] e^{-2r/a_0}$$

Now at $r_{max} = a_0$

$$\left[\left(\frac{-2}{a_0}\right) \left\{ \left(\frac{-2}{a_0}\right) a_0^2 + 4a_0 \right\} + 2 \right] e^{-2 a_0/a_0}$$

brought back from Eq. 1 $= * \left(\frac{4}{a_0^3}\right)$

$$= (+4 - 8 + 2) \times e^{-2} < 0$$

$$= -2 \times e^{-2}$$

So, $\frac{d^2P(r)}{dr} < 0$

therefore a maximum !